CRICKETLENS | DEVELOPMENT REPORT

Session 01 - Repo foundation and data-processing starter | 17 October 2026 | Status: in progress

Purpose

Begin implementation of an EDA-first IPL analytics platform with six dashboard tabs, two probability models, and a no-login prediction tracker. Maintain clear evidence of what has been built, tested, and not yet completed.

Finalized project design

Component	Agreed functionality
Tabs 1-4	IPL Overview; Team Analysis; Player Explorer; Statistics & EDA Lab
Tab 5	Prediction Arena: batter 30+/50+ milestone probabilities plus wicket in the next over (six legal deliveries)
Tab 6	Prediction Tracker: save every forecast and actual outcome; evaluate probability quality; no automatic training
ML lifecycle	Develop through 2024, validate on 2025; refit on data through 2025, hold out 2026; then refit through 2026 for future matches
Website	No login or signup; browser-local history initially; historical replay later

Delivered in this session

Item	Evidence / status
Python project scaffold	README, pyproject, gitignore, modular source files, tests and documentation
Cricsheet JSON parser	Match/delivery/batter innings/next-over checkpoint CSV extraction coded
First wicket baseline	Logistic regression code, constant-rate comparison and time-based validation; no real data trained
EDA starter	Basic match/run/wicket/milestone descriptive summary generator
Engineering test	4 automated tests passed, using synthetic match data and synthetic season split

What the test does and does not show

The synthetic fixture verifies wide/no-ball legal-delivery accounting, team wicket labels, franchise renames, and checkpoint construction. Passing four tests confirms starter-code behavior on those cases only; it is NOT a claim about performance on the full IPL data or about predictive accuracy.

CRICKETLENS | NEXT IMPLEMENTATION STEPS

Work plan aligned to the IT462 Exploratory Data Analysis syllabus

Data and academic analysis

Stage	Next concrete work and report output
Data acquisition	Place official Cricsheet IPL JSON ZIP in data/raw, verify actual season coverage and record source credit.
Stage II screening	Audit file count, bad/missing fields, unusual innings, duplicate IDs, censored overs; document cleansing decisions.
EDA notebook	Generate distributions, five-number summaries, variability, correlations and team/player/venue comparisons; interpret genuine measurements.
Advanced EDA	Document sampling bias and missingness mechanisms; add simple regression, multicollinearity and season trends.

Model development sequence

Model	Next task
Next-over wicket	Run baseline on real historical over checkpoints; compare validation Brier/log loss with the constant-rate baseline; examine calibration.
Batter milestones	Build pre-match rolling form without leakage; handle non-participation/DNB; enforce P(50+) <= P(30+).
Holdout and future	Keep 2026 unseen while choosing models. After one unbiased 2026 evaluation, build separate refitted models for 2027.

Current blockers and integrity controls

Data availability: The user's earlier IPL ZIP is not mounted here, and direct official retrieval was unsuccessful. No real IPL records or test metrics were created. The full dataset must be uploaded or downloaded locally before actual training begins.

GitHub: A private repository named CricketLens is recommended. The starter project is ready for a first commit, but no remote GitHub repository has been created or pushed from this workspace.

Report policy: Update docs/DEV_LOG.md after each coding session and reports/REPORT_DRAFT.md as actual data checks, figures, model evaluation and findings become available. Cite Cricsheet and disclose AI assistance according to the course policy.

Immediate next action

Upload **ipl_json.zip** or put it in **data/raw/**, execute the parsing command in README.md, and review the resulting screening issues and season coverage before training.